

Niraj Chavan

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Summary

Computer Engineering student with strong problem-solving skills and expertise in C++, Data Structures & Algorithms, AI/ML, and software development. Passionate about building scalable software and intelligent systems to solve real-world challenges.

EDUCATION

Vishwakarma Institute of Technology Pune, India

B.Tech, Computer Science Engineering — CGPA: 9.09/10 (Pre-Final Year), consistently in top academic bracket 2024 –2028

PROJECTS

Federated Learning — Privacy-Preserving Autism Detection



Tech Stack: C++, Python, TensorFlow, Keras, MobileNetV2, EfficientNetB4, OpenCV, Flower, Federated Learning.

- Designed a **privacy-preserving federated learning framework** for autism detection through facial analysis, enabling collaborative model training across distributed medical datasets **without centralizing sensitive patient data**.
- Implemented **MobileNetV2** and **EfficientNetB4** models with optimized preprocessing and federated model aggregation, achieving **90%** and **85% classification accuracy**, respectively, on cross-cultural autism datasets.
- Developed a **GDPR-compliant** multi-site diagnostic pipeline using secure client-server communication and decentralized training, **eliminating raw-data sharing** while preserving model performance.
- Evaluated multiple deep learning architectures and federated training strategies across heterogeneous client datasets, demonstrating the feasibility of **scalable privacy-preserving AI for medical diagnosis**.

SeeTheScience — AI-Powered Physics & Chemistry Simulator



Tech Stack: C++, Python, Node.js, MongoDB, Gemini API, OpenAI API, Anthropic API, Groq API, HTML, CSS, JavaScript

- Developed** an AI-powered educational platform that converts natural-language physics and chemistry problems into **23 interactive 2D simulations** with step-by-step KaTeX mathematical derivations, making complex concepts easier to visualize and understand.
- Integrated** four LLM providers (OpenAI, Gemini, Anthropic, and Groq) with an intelligent auto-failover mechanism and implemented an offline regex-based parser, **ensuring uninterrupted simulation generation even during API outages**.
- Engineered** a real-time visualization system featuring Recharts analytics, dynamic variable manipulation, and a drag-and-drop Scene Builder, **enhancing user interaction and experimentation with scientific concepts**.
- Designed** a scalable modular architecture combining AI services, simulation rendering, and quiz generation, **delivering an extensible platform capable of supporting future educational content and simulation modules**.

Driver Drowsiness Detection System



Tech Stack: Python, Flask, TensorFlow/Keras, ResNet50V2, OpenCV, JavaScript, HTML, Tailwind CSS

- Developed** a real-time driver drowsiness detection system using the **ResNet50V2** deep learning model to classify eye states (open/closed) from live webcam streams, enabling **proactive fatigue alerts** for improved road safety.
- Implemented** a computer vision pipeline using **OpenCV** for face and eye detection with frame-by-frame image preprocessing, **providing reliable real-time eye-state classification under continuous video input**.
- Engineered** an end-to-end inference workflow integrating camera capture, image preprocessing, deep learning inference, and alert generation, **delivering low-latency predictions suitable for real-time applications**.
- Designed and deployed** a responsive **Flask** web application with seamless JavaScript frontend-backend communication and a **Tailwind CSS** interface, **enabling cross-platform accessibility through any modern web browser**.

ACHIEVEMENTS

1st Runner-Up, Hacktivate — CODEFLIX'26 | CSI VIT Pune — top placement in institute-level competitive hackathon.

Strong foundation in Data Structures, Algorithms, and competitive programming with proven ability to solve complex algorithmic and optimization problems efficiently.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, SQL.

ML & AI: TensorFlow, Keras, Scikit-learn, OpenCV, MobileNetV2, EfficientNetB4, ResNet50V2, Federated Learning.

Web & Backend: Node.js, Express.js, Flask, HTML5, CSS3, JavaScript, REST APIs, MongoDB, MySQL, PostgreSQL, PL/SQL.

Core CS: DSA, OOP, DBMS, OS.

Tools: Git, GitHub, Linux, VS Code, Google Colab.